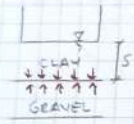
• $Q = k \cdot i \cdot A \rightarrow Q = k \cdot \frac{\Delta h}{L} \cdot A$ → 1 m² (per unit area)

$Q = (2\times10^{-8}) \cdot \frac{(7-5)}{4} \cdot (1) = 1\times10^{-8}\ m^3/s/m^2 = 0{,}315\ m^3/years/m^2$

• $H_{seepage(max)} = ?$



$i_{cr} = \frac{\gamma_A}{\gamma_w} = \frac{(18-9{,}81)}{9{,}81} = 0{,}835$

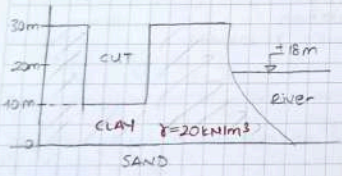
$i = \frac{7-5}{5} =$

$FS = 1{,}5 = \frac{0{,}835}{i} \rightarrow \frac{0{,}835}{1{,}5} = \frac{7-5}{S}$

$S = 4{,}5\ m$

$H = 48 - (40 + 4{,}5) \rightarrow H = 3{,}5\ m$

Question 4)



FS = ?

$\sigma_0 = (\gamma_{sat}) \cdot z$

$= 20 \cdot 10 = 200\ kPa \downarrow$

$\sigma_w = u \cdot \gamma_w$

$\sigma_w = (18) \cdot (9{,}81) = 176{,}6\ kPa \uparrow$

$FS = \frac{200}{176{,}6} = 1{,}13$

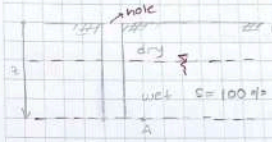
yapıray

STRESSES in SOILS

Stresses in soils can be subdivided into two groups

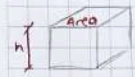
Stresses caused by self-weight of soil layers (σ, σ', u)

Stresses caused by either external loadings such as buildings, bridges, highways dams ect. and/or lowering of GWT ($\Delta\sigma$)



$W = \frac{\rho \cdot g \cdot V}{\gamma}$ ρ: density; g: acceleration due to gravity; V: volume; W = Force

$W = \gamma \cdot V = \gamma \cdot A \cdot h = \gamma \cdot A \cdot z$ (V = A × h: volume, area, depth)



$Force_{Total} = Force_{\substack{solids\\grains+\\particles}} + Force_{water}$

$\frac{\gamma \cdot \cancel{A} \cdot z}{\cancel{A}} = \frac{Force_{solids}}{A} + \frac{\gamma_w \cdot \cancel{A} \cdot z_w}{\cancel{A}}$

$\Rightarrow \gamma \cdot z = \frac{Force_{solids}}{A} + \gamma_w \cdot z_w$

$\sigma \qquad \sigma' \qquad u \rightarrow \sigma = \sigma' + u$

σ = total stress → the self weight of soil layers above the point of interest

σ' = effective stress → (carried by soil grains through contact between particles)

u = pore water pressure (carried by water)

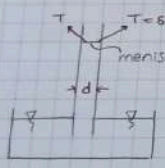
yapıray

for dynamic conditions (with flow)

$h = \frac{v^2}{2g} + \frac{u}{\gamma_w} + z$ → from that $u = (h - z)\cdot\gamma_w$

$h_c = \frac{4 \cdot T}{d \cdot \gamma_w}$ — $T \cong 0.073$ N/m at a room temperature, $10°C - 20°C$, $\gamma_w \cong 10$ kN/m³

$h_c\,(m) \cong \frac{0.00003}{d\,(m)}$ — h_c = capillary rise ≅ height of capillary rise ≅ height of capillary water



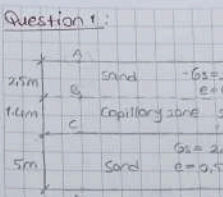
for static conditions (with no flow)

$u = \gamma_w \cdot z_w$

$\sigma_v = \sigma_v' + u \qquad \sigma_h = \sigma_h' + u$

$K_0 = \frac{\sigma_h'}{\sigma_v'} \Rightarrow K_0$ = lateral earth pressure at rest coefficient $= 1 - \sin\phi$

Question 1:



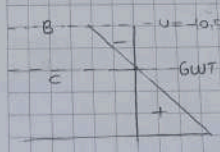
Calculate the σ_v' for A, B, C, D. Draw the σ_v', σ_v, u with depth

$\gamma_{dry(sand)} = \frac{G_s \cdot \gamma_w}{1+e} = \frac{2.67 \cdot 9.81}{1+0.5} = 17.46$ kN/m³

$\gamma_{n(sand)} = \frac{\gamma_w (G_s + e\,S)}{1+e} = \frac{(9.81)(2.67 + 0.5 \cdot 0.50)}{1+0.5} = 19.10$ kN/m³

$\gamma_{sat(sand)} = \frac{\gamma_w (G_s + e\,S)}{1+e} = \frac{(9.81)(2.67 + 0.5 \cdot 1)}{1+0.5} = 20.73$ kN/m³

Point A	Point B
$\sigma_v = 0$	$\sigma = \gamma_{dry} \cdot z = (17.46)\,2.5 + 43.65$ kN/m²
$u = 0$	$u = -(0.5)(9.81)(1.4) = +6.87$ kN/m²
$\sigma_v' = 0$	$\sigma' = (43.65) - (-6.87) = 50.52$ kN/m²



yapıray

yapıray

Point C

$u_C = 0$

$\sigma_C = \sigma_B + (1.u)(19.10) = 70.39\ kN/m^2$

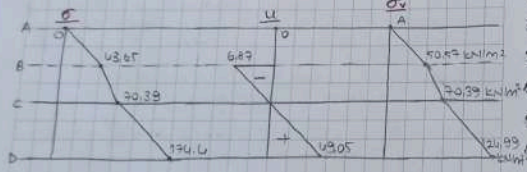
$\sigma'_C = 70.39\ kN/m^2$

Point D

$\sigma = \sigma_C + 5.(20.3) = 174.0u\ kN/m^2$

$u_D = 5.19.81 = u9.05\ kN/m^2$

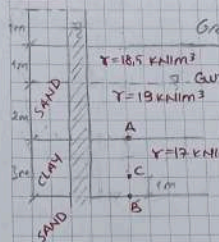
$\sigma'_D = 174.0u - 48.05 = 12u.99\ kN/m^2$



Question 2:

There is a artesian pressure at the bottom of the soil profile (in the sand layer). The water rises 3m in the piezometer that is inserted to the sand layer. Determine;
- Total stress
- Effective stress
- Pore water pressure

acting on A,B,C points



Point A

$\sigma_A = 1.18,5 + 2.19 = 56,5\ kN/m^2$

$u_A = 2.9,81 = 19,6\ kN/m^2$

$\sigma'_A = 56,5 - 19,6 = 36,9\ kN/m^2$

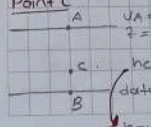
Point B

$\sigma_B = \sigma_A + 3.17 = 107,5\ kN/m^2$

$u_B = 4.9,81 = 68,7\ kN/m^2$

$\sigma'_B = 107,5 - 68,7 = 38,8\ kN/m^2$

Point C



$u_A = 2m$, $z = 3m$ → $h_A = u + z = 5m$

$h_C = 3 - 1.(0,67) = 6.33m$

datum: $z = 0$, $u_B = 3m$ → $h_B = u + z = 3m$

$\Delta h = 2m$

$i = \frac{\Delta h}{L} = \frac{2}{3} = 0.67$

$h = u + z$

$6.33 = u + 1 \rightarrow u = 5,33\ m$

yapıray

yapıray

Point C

$u_C = 5.33 \cdot 9.81 = 52.3\ kN/m^2$

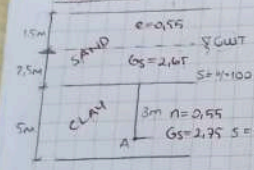
$\sigma_C = \sigma_A + 2.19 = 90.5\ kN/m^2$

$\sigma'_C = 90.5 - 52.3 = 38.2\ kN/m^2$

Question 3:

a) Calculate the total and effective pressure at point A

b) Determine the change in effective stress at point A if the GWT is lowered 2m. (Assume that the soil is located above the GWT is in dry condition)



a) $\gamma_{dry,sand} = \frac{G_s \cdot \gamma_w}{1+e} = \frac{2,65 \cdot 9,81}{1+0,55} = 16,77\ kN/m^3$

$\gamma_{sat,sand} = \frac{(G_s+e)\gamma_w}{1+e} = \frac{(2,65+0,55)\cdot 9,81}{1+0,55} = 20,25\ kN/m^3$

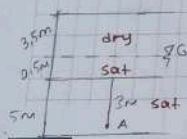
$n = \frac{e}{1+e} \rightarrow 0,55 = \frac{e}{1+e} \rightarrow e = 1,22$

$\gamma_{sat,clay} = \frac{(G_s+e)\gamma_w}{1+e} = \frac{(2,75+1,22)\cdot 9,81}{1+1,22} = 17,54\ kN/m^3$

$\sigma_A = (1,5)(16,77) + (2,5)(20,25) + (3)(17,54) = 128,40\ kN/m^2$

$u_A = (5,5)(9,81) = 53,9\ kN/m^2$

$\sigma'_A = 128,4 - 53,9 = 74,45\ kN/m^2$



$\sigma_A = 3,5 \cdot 16,77 + 0,5 \cdot 20,25 + 3 \cdot 17,54 = 121,40\ kN/m^2$

$u_A = 3,5 \cdot 9,81 = 34,34\ kN/m^2$

$\sigma'_A = 121,40 - 34,34 = 87,105\ kN/m^2$

$\Delta\sigma'_A = 87,105 - 74,45 = 12,66\ kN/m^2$

yapıray

① The test results on a soil sample [m = 1800g and v = 1000 cm³] taken from a construction site were presented below. Calculate w (water content), e (void ratio), n (porosity), S (degree of saturation), ρ_s (density of solids), ρ_{dry} (dry density of soil), ρ_n (moist density of soil), ρ_{sat}, γ_s (unit weight of solids), γ_{dry} (dry unit weight of soil), γ_n (moist unit weight of solid), γ_{sat}

$M_{dry} = 1500g$ (after drying in an oven at 105°C)

$G_s = 2,7$ (specific gravity) $= \frac{\rho_s}{\rho_w} = \frac{\rho_s \cdot g}{\rho_w \cdot g} \Rightarrow G_s = 2,7 = \frac{\rho_s}{1\ g/cm^3} \Rightarrow 2,7\ g/cm^3$, $\rho_w = 1\ g/cm^3$

$w = \frac{M_w}{M_{solid}} = \frac{M_n - M_{dry}}{M_{dry}} = \frac{1800-1500}{1500} = \frac{300}{1500} = 0,2 \quad 20\%$

$e = \frac{V_{voids}}{V_{solids}} = \frac{V_v}{V_s} = \frac{V_a + V_w}{V_s} = \frac{(144,45+300)}{555,55} = 0,8$, $n = \frac{V_v}{V_t} = \frac{144,45+300}{1000} \cong 0,45$

$S = \frac{V_w}{V_v} \cdot 100 = \frac{300}{[144,45+300]} \cdot 100 = 67,5\%\ (0,675)$

$V = 1000\ cm^3 = V_a + V_w + V_s = V_a + 300 + V_s \rightarrow V_a = 144,45\ cm^3$

$\rho_s = \frac{M_s}{V_s} \Rightarrow 2,7\ g/cm^3 = \frac{1500g}{V_s} \Rightarrow V_s = 555,55\ cm^3$

$\rho_w = \frac{M_w}{V_w} \Rightarrow 1,0\ g/cm^3 = \frac{300g}{V_w} \Rightarrow V_w = 300\ cm^3$

$S \cdot e \cdot \rho_w = w \cdot \rho_w$ or $S \cdot e \cdot \gamma_w = w \cdot \gamma_s$

$2,675 \cdot e \cdot 1 = 0,2 \cdot 2,7 \Rightarrow e = 0,8$

$S \cdot 0,8 \cdot 1 = 0,2 \cdot 2,7 \Rightarrow S = 0,675$

$\rho_{dry} = \frac{M_{dry}}{V_t} = \frac{1500g}{1000\ cm^3} = 1,5\ g/cm^3$, $\rho_{dry} = \frac{\rho_s}{1+e} = \frac{2,7}{1+0,8} = 1,5\ g/cm^3$

$\rho_n = \rho_n = \frac{M}{1000\ cm^3} = \frac{1800}{1000} = 1,8\ g/cm^3$, $\rho_n = \frac{\rho_s + S \cdot e \cdot \rho_w}{1+e} = \frac{2,7 + 0,675 \cdot 0,8 \cdot 1}{1+0,8} = 1,8\ g/cm^3$

$\rho_{sat} = \frac{M}{V} = \frac{1800 + 144,45}{1000} = 1,95\ g/cm^3$

$\rho_{sat} = \frac{\rho_s + S \cdot e \cdot \rho_w}{1+e} = \frac{2,7 + 1 \cdot 0,8 \cdot 1}{1+0,8} = 1,95\ g/cm^3$

yapıray

$\gamma_s = \rho_s \cdot g = 2.7 \cdot 10 = 27\ kN/m^3$

$G_s = \frac{\gamma_s}{\gamma_w} \Rightarrow \gamma_s = G_s \cdot \gamma_w = 2.7 \cdot 10 = 27\ kN/m^3$

$\gamma_{dry} = \rho_{dry} \cdot g = 1.5 \cdot 10 = 15\ kN/m^3$

$\gamma_n = \rho_n \cdot g = 1.8 \cdot 10 = 18\ kN/m^3$

$\gamma_{sat} = \rho_{sat} \cdot g = 1.95 \cdot 10 = 19.5\ kN/m^3$

$\gamma_{dry} = \frac{\gamma_s + S \cdot e \cdot \gamma_w}{1+e} = \frac{2.7}{1.8} = 15\ kN/m^3$

$\gamma_n = \frac{\gamma_s + S \cdot e \cdot \gamma_w}{1+e} = \frac{27 + 10 \cdot 0.695 \cdot 0.8}{1+0.8} = 18\ kN/m^3$

$\gamma_{sat} = \frac{\gamma_s + S \cdot e \cdot \gamma_w}{1+e} = \frac{27 + 1.0 \cdot 0.8 \cdot 10}{1+0.8} = 19.5\ kN/m^3$

② Classify the soils according to USCS and AASH to Atterberg limits.

	SOIL A	SOIL B	SOIL C
Sieve No	Passing %	Passing %	Passing %
4	100	98	100
10	90	90	90
40	80	80	80
200	4	10	52
LL (%)	—	24	55
PL	—	16	29
Cu	8.5	7	—
Cc	1.7	1.5	—

SOIL A

0 % gravel
4 % fine-grained (silt + clay) → hydrometer test
96 % sand

No 200 → 4 % < 5 % → SW, SP

$C_u = 3.5 < 6$

$C_c = 1.7\ [1-3]$ A-3 → N.P

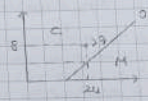
(gravel and sand)

Sieve analysis is used to seperate coarse grained soils from fine grained soils (silt and clay). Hydrometer test is used to seperate silt from clay.

SOIL B

2 % gravel
10 % silt + clay
88 % sand

SW - SC



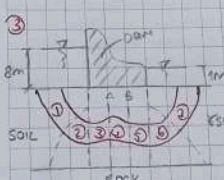
SOIL C

0 % gravel — 55 > 50 → H
48 % sand — 0.73 (55-20) = 25.50 → Ip = PI Atine
52 % clay — Ip = PI given = 55-29 = 26

CH

LL-30 = 55-30 = 25 < 26 → A-7-6

③



$\gamma_{sat} = 20 kN/m^3$

From a constant head permeameter test ($\Delta h = 40$ cm, $\Delta t = 120$ s) on a soil sample ($L = 20$ cm, $A = 30\ cm^2$) taken from a foundation soil under a dam shown in the figure, Q measured as 250 cm^3 ($\ell = 2m$)

a) Calculate q (flow rate; seepage rate) under the dam
b) u_A and u_B
c) Check the stability at exit points

yapıray

yapıray

a) $q = k \cdot \Delta H \cdot \frac{N_f}{N_d}$ $\quad k = \frac{Q \cdot L}{A \cdot \Delta h \cdot \Delta t} = \frac{250 \cdot 20}{30 \cdot 40 \cdot 120} = 3.5 \times 10^{-2}$ cm/s $= 3.5 \times 10^{-4}$ m/s

$q = [3.5 \times 10^{-4}] \cdot [8-1] \cdot \frac{3}{7} \cong 1.4 \times 10^{-3}$ m³/s/m

b) $h_A = \frac{u_A}{\gamma_w} + z_A$ $\quad h_I = h_{II} = 8$m

$h_A = n_I - (\text{numbers of steps}) \cdot \frac{1}{3}$ (loss of energy at each point) $= 5$m

$h_I = 0 + 8 = 8$m
$h_{II} = 6 + 0 = 6$m
$h_{III} = 1 + 0 = 1$m
$h_{IV} = 0 + 1 = 1$m

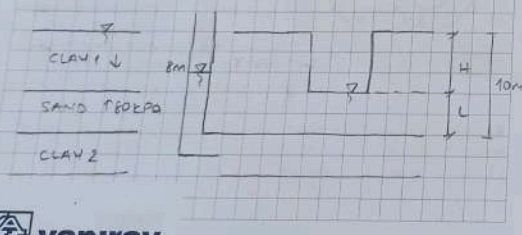
$u_A = 50$ kPa

$h_B = \frac{u_B}{\gamma_w} + z_B \rightarrow h_I - u.l = um \rightarrow 4m = \frac{u_B}{10} + 0 \rightarrow u_B = 40$ kPa

c) i exit must be compared with $i_e = i_{cr}$

$i_{cr} = \frac{\gamma_A}{\gamma_w} \cong 1 \Rightarrow \frac{\gamma_{sat} - \gamma_w}{\gamma_w} = \frac{20}{10} - \frac{10}{10} = 1$ $\quad i = \frac{\Delta h}{L} = \frac{1}{2} = 0.5 < i_{cr} = 1$

④ At a construction site, an excation is to be made in a clay 1 layer. Before the excavation, water level was risen by 6m in a piezometer tube inserted down to the sand layer. Calculate the depth of excavation which will cause the cut to fail.



$L.20 = 50$ kPa $\rightarrow L = 4$m

$i_{cr} = i = \frac{\gamma_{sat} - \gamma_w}{\gamma_w} = \frac{20-10}{10} = 1 = \frac{\Delta h}{L} = \frac{6-L}{L} = 1 \rightarrow 6 - L = L$ $\quad L = 4$m

STRESS INCREMENTS IN SOILS

Increment in stresses within a soil layer is caused by either an external load and/or lowering of the level of GWT (ground water table) In order to calculate the value of $\Delta\sigma$, two methods are generally chosen. These methods are: using simple curves, using a simplified procedure

In using some simple curves, the geometry of loaded are is highly important to calculate the value of $\Delta\sigma$ Loaded area can be
- square etc.
- rectangle
- round

In using the simplified procedure, a simple equation is enough to calculate the value of $\Delta\sigma_z$

If the loaded area is a square or a rectangle and simple curves is preferred as the method for calculating the value of $\Delta\sigma_z$ the following procedure can be followed;

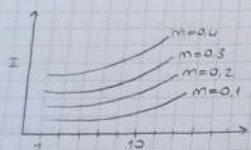
- calculate m by $\frac{x}{z}$
- calculate n by $\frac{y}{z}$ } the depth of interest below one of the corners of the loaded area
- calculate I by m and n from some curves
- calculate $\Delta\sigma_z$ by $q_o \cdot I \rightarrow$ (no unit)

($\Delta\sigma_z$) (kN/m²) (kPa) $\quad$ (kN/m²) (kPa)

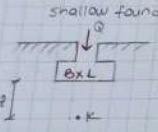
yapıray
yapıray

x: short dimension of the loaded area
y: the long " " " " "
I: influence factor
q_o: the load on the base of the loaded area
z: the depth of interest

Note! m and n are actually interchangeable



shallow foundation

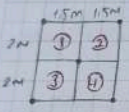


$q_o = \frac{Q}{B \times L}$

A cross-section



A bird's eye view



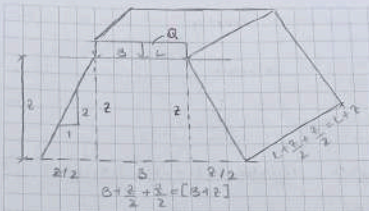
$m = \frac{B}{z} = \frac{1.5m}{10m} = 0.15$

$n = \frac{L}{z} = \frac{2m}{10m} = 0.2m \rightarrow I_1, I_2, I_3, I_4$

$\Delta\sigma = \sigma_z = \sigma_{1,3} = q_o \left[\frac{Q\,kN}{B \times L} \right] \left[I_1 + I_2 + I_3 + I_4 \right]$

$4I$

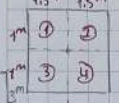
$\sigma_v = \sigma'_v + u + \Delta\sigma$



In using the simplified procedure, a simple equation is enough to calculate the value of $\Delta\sigma_z$. For this method, $\Delta\sigma$ is calculated by the following equation.

$$\Delta\sigma = \frac{Q\,(kN)}{(B+z)(L+z)}$$
(kPa) (m) (m)

Question1: Q=1200 kN load is acting uniformly on a shallow foundation with 2m x 3m size. Determine the stress increment at the center of the foundation for depth 3m, 6m, 10m according to Boussinesq and simplified method.



z(m)	Q(kN)	m=B/z	n=L/z	I (-)
3m	1200	1/3=0.33	1.5/3=0.5	0,062
6m	1200	1/6=0,16	1.5/6=0.25	0,019
10m	1200	1/10=0,1	1,5/10=0,15	0,05

$I_1 = I_2 = I_3 = I_4$

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$\Delta\sigma_z$ (kPa)

(1) (0.067) (200) = 49.8
(4) (0.021) (200) = 15,2
(4) (0.005) (200) = 4

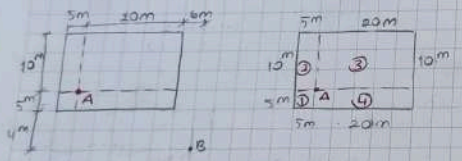
decrease

$\Delta\sigma_z = q_0 . I$ → $\frac{1200}{2.3} = 200$ kPa

$\Delta\sigma_z = q\left[\frac{B \times L}{(B+z)(L+z)}\right]$ or $\frac{Q(kN)}{(B+z)(L+z)}$

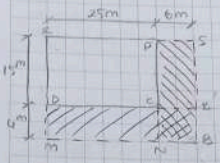
z(m)	Q(kN)	(B+z)m	(L+z)m	$\Delta\sigma_z$(kPa)
3	1200	5	6	40
6	1200	8	9	16.67
10	1200	12	13	7.69

Question 2: A shallow foundation (15m x 25m) is loaded 120 kPa uniformly. Determine the stress increase at point A for 10m depth and at point B for 5m depth.



Area	m=B/z	n=L/z	I
1	5/10 = 0,5	5/10 = 0,5	0,084
2	20/10 = 2	5/10 = 0,5	0,185
3	20/10 = 2	10/10 = 1	0,2
4	10/10 = 1	5/10 = 0,5	0,1202
			+
			0,5392

$\Delta\sigma_z = I\ q_0 = (0,5392)(120) = 64.7$ kPa



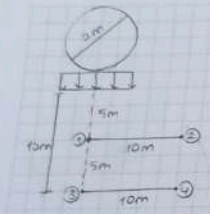
Area	m=B/z	n=L/z	I
BSEM	31/5 = 6,2	18/5 = 3,6	0,2464
BKDM	31/5 = 6,2	4/5 = 0,8	0,1849
BSFN	6/5 = 1,2	13/5 = 3,6	0,2163
GNICK	6/5 = 1,2	4/5 = 0,8	0,1654

$I_{RCDA} = I_{BSEM} - I_{BKDM} - I_{BSFN} + I_{GNCC}$
$I_{RCDA} = 0,2464 - 0,1849 - 0,2163 + 0,1654 = 0,0136$
$\Delta\sigma_z = I_{total} \cdot q_0 = (0,0136)(120) = 1.63$ kPa

Question 3: Q=1000 kN load is acting uniformly from oil tank to the circular shallow foundation. The diameter of the foundation is 2m. According to the information given above: Find the stress increament at the center and the 10m horizontally far from the center of the foundation for the depth 5m and 10m with Boussinesq method.

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$\Delta\sigma_z = \frac{I \cdot q_0}{100}$ $q_0 = 1000/\frac{\pi \cdot z^2}{4} = 318,31\,kPa$

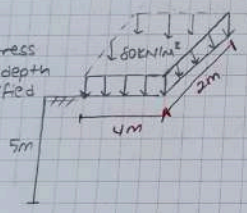


Point	z(m)	r(m)	x(m)	z/r(m)	x/r(m)	I
1	5	1	0	5	0	6
2	5	1	10	5	10	0,12
3	10	1	0	10	0	1,4
4	10	1	10	10	10	0,28

q_0	$\Delta\sigma_z$(kPa)
318,31	(6×318,31)/100 = 19,1
318,31	4,46
318,31	0,38
318,31	0,8

Question 4:

Determine the stress increase at 5 m depth according to simplified method.



$\Delta\sigma_z = \frac{Q}{(B+z)(L+z)}$ or $\frac{q\,(B\times L)}{(B+z)(L+z)}$

$\Delta\sigma_z = \frac{60(4\times2)}{(4+5)(2+5)} = 10,16\,kPa$

SETTLEMENT of SOILS

Soils get compressed under the effect of loads. The compression of soils is subdivided into two categories:
① consolidation ② compaction

∝ Consolidation: the most important parameter is the presence of water in voids.
∝ compaction: the most important parameter is the presence of air in voids.

Consolidation

The settlement caused by the effect of water is subdivided into 4 categories. They are:
1) immediate settlement (volume decreases, effective stress increases)
2) consolidation settlement (primary settlement decreases effective stress increases)
3) secondary settlement (volume is constant, effective stress increases)
4) creep-related settlement (volume and effective shear stress are constant)

in other words, the total settlement can be written in the following form:

$$S_t = S_i + S_c + S_s + S_{creep}$$

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In sands [S_i + S_c] form the largest part of S_t (total settlement). [S_s + Screep] can be ignored. Because the value of c is high and therefore the process of settlement is assumed to keep pace the construction. In sands, due to the above explanation, S_i and S_c may not be distinguished. The value of settlement of a sand on a sand-like soil can be calculated by using such an equation

$$S = q \frac{B}{N^{0.6}}$$

- q = base pressure (kPa)
- N = SPT number (Standard Penetration Test)
- B = the short dimension of the foundation

In addition another equation proposed by Schmetmann can be also used to calculate the value of s for sand soils.

$$S = C_1 \cdot C_2 (q_1 - q_2) \sum_{i=1}^{n} \frac{I}{E} \cdot \Delta z$$

C_1: coefficient; C_2: coefficient; $(q_1 - q_2)$: pressures; I: influence factor; Δz: thickness of sublayers of sand soil; E: modulus of elasticity

In inorganic clays the main part in other words the largest part of S_t is the value of S_c. However S_i can be also calculated in addition to S_c by using such an equation below

$$S_i = q \frac{B(1-\nu^2) I}{E}$$

q: base pressure $q = \frac{Q}{B \times L}$

ν: poisson's ratio [0,3 - 0,5 for soils; for silts = 0,33 (If clay is completely saturated $\nu = 0,5$) clays = 0,4 - 0,5

I: influence factor which is dependent on the ratio of L/B. But generally it is between 0,7 - 1,2

E: modulus of elasticity [250 cu - 1000 cu] for clays

Cu: undrained cohesion

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SPT number; cone tip resistance core point

Note: $E = 766 N$ (for sands) or $2,5 - 4,5\ q_c$ (for sands)

In inorganic clays the amount of consolidation is dependent on the presence of the water in soils. According to the consolidation theory proposed by Tergeyni, for consolidation soil must be completely saturated. Actually compression of soils may be conceivably attributed to

- by compression of soils (particles, grains; if it is not true grains are assumed to be incompressible.)
- by compression of water (if it is not true water is assumed to be incompressible.)

- By the expulsion, escape, flow, of water in the voids out of a soilmass so, the particles get closer to flow it is true.
- Encholter resulting in a decrease in volume which is known as settlement as a matter of fact settlement of consolidation which is known as primary settlement
- consolidation is a type of compression settlement if it is dependent on the escape / flow / expulsion of water out of soils
- The process of escape / expulsion / flow of water will occur according to Darcy's Law ($v = k \cdot i$)

The main factor in the process of consolidation is the movement of water

The main reason of

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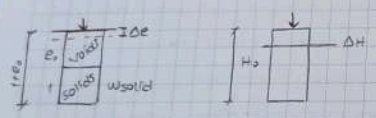
Sc can be calculated by using the following equations;

$$\frac{\Delta H = Sc}{H_0} = \frac{\Delta e}{1+e_0}$$

$$\Delta H = Sc = \frac{C_c}{1+e_0} H_0 \log\left[\frac{\sigma_{v0}+\Delta\sigma}{\sigma_{v0}}\right]$$

for normally consolidated clays $[\sigma_p = \sigma_{v0}]$

$$\Delta H = Sc = m_v H_0 \Delta\sigma$$



$V_{solid} = 1$ $\quad \frac{\Delta e}{1+e_0} = \frac{\Delta H}{H_0}$

$$e = \frac{V_{voids}}{\underbrace{V_{solids}}_{1}}$$

Consolidation will occur either by an external loading and/or by lowering the GWT and the self weight of the soil layers. For external loading: when a soil layer is subjected to an external loading such as a building, a dam, a bridge etc., an increase in the stresses within that soil layer will occur. This increase is designated by $\Delta\sigma$. The $\Delta\sigma$ is first carried by water in the voids of that soil layer. Therefore, at first, pore water pressure will increase and the value of it will be $u_0 + \Delta u$.

$\sigma_v = \sigma_{v0}' + u_0$ (just before an external load is applied on that soil layer) $\quad t = 0$

$(\sigma_v' + \Delta\sigma) = \sigma_{v0}' + (u_0 + \Delta\sigma)$ (just after an external load is applied on that soil layer) $\quad t = t_1$

Note: at $t = t_1$, $\Delta\sigma = \Delta u$

In time, with draining of water out of that soil layer, Δu will start to decrease and correspondingly the increment ($\Delta\sigma$) in stresses caused by an external load will start to be transferred to soil skeleton (grains).

Note: Δu is referred to as excess pore water pressure (in kPa)

$$\sigma_v' + \Delta\sigma = [\sigma_{v0}' + \Delta\sigma_1'] + [u_0 + \Delta\sigma_2] \quad t = t_2$$

Note: $\Delta\sigma_1' + \Delta\sigma_2 = \Delta\sigma$

When all the increment ($\Delta\sigma$) in stresses are transferred to soil skeleton from water, the process of consolidation is completed.

In other words, when excess pore water pressure (Δu) is completely dissipated, the process of consolidation is completed.

In other words, when the value of pore water pressure $[u_0 + \Delta u]$ returns to the value $[u_0]$ just before the external load is applied, the process of consolidation is completed.

For the lowering of ground water table,
- when the GWT is lowered, u_0 will certainly decrease $[\sigma_v = \sigma_{v0}' + u_0]$
- If u_0 decreased by this process, σ_{v0}' will certainly increase $[\sigma_v' > \sigma_{v0}']$

↙ after the lowering of GWT ↘ before the lowering of GWT

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- This difference between σ_2' and σ_{v0}' will form the value of $\Delta\sigma$.
And, due to this $\Delta\sigma$, consolidation will occur in that soil layer.

Note! $\Delta\sigma$ is calculated by either using the simplified procedure [2:1 method] $\rightarrow \Delta\sigma = \frac{Q}{(B+z)(L+z)}$

or using the same simple curves for determining the value of $I \rightarrow \Delta\sigma = q_0 \cdot I$

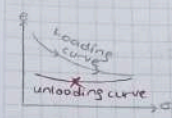
Oedometer Test

The amount of s_c (consolidation settlement)
The value of a_v (coefficient of compressibility)
The value of m_v (coefficient of volumetric compressibility)
The value of C_c (compression index)
The value of C_v (coefficient of consolidation)

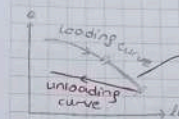
these are calculated by a laboratory test which is called oedometer test.

In this test, a soil sample (more or less 50 mm in diameter and 20 mm thick) is placed in a circular steel ring. Two porous stones are placed on the of this sample and below this sample. The consolidation cell is filled with water. After that, several vertical loads are applied on this soil sample. The usual sequence of loading is 25, 50, 100, 200, 400 and 800 kPa. Each load is kept constant on the soil sample for 24 hours. After the last load is applied on the sample for 24 hours, the load is decreased 1/4 or 1/2 the value of the last load and at this stage, the process is lasted for 24 hours.

In the end of such a test, the following graphs can be constructed:

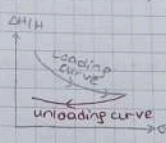


→ the slope of black line will give us $a_v = \frac{\Delta e}{\Delta\sigma'}$



the slope of the straight parton of the black line will give us the value of C_c

$C_c = \frac{\Delta e}{\Delta \log\sigma'}$

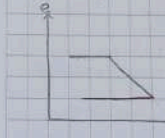


→ the slope of black one will give us $m_v = \frac{\Delta H/H}{\Delta\sigma'}$

Note! e: void ratio, σ': effective stress (vertical load applied)

Note! e - logσ' plot is a semi logarithmic graph.

Note! For practical purposes,



this graph can be shown like that

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Normally Consolidated and Over Consolidated Soils

If the past vertical pressure / past max effective stress (σ_p') / past vertical stress to which any soil layer has been ever subjected is greater than the present vertical pressure / existing effective stress (σ_{vo}') / present over burden stress ... then the soil is overconsolidated
[if $\sigma_p' > \sigma_{vo}'$; soil is overconsolidated]

Note: most of 80 % of soils in nature is overconsolidated.

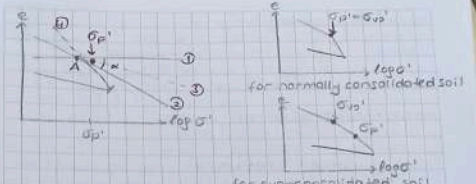
If σ_p' is equal more or less to σ_{vo}', then soil is normally consolidated [if $\sigma_p' = \sigma_{vo}'$ soil is normally consolidated]

σ_p' is also known as preconsolidation pressure. σ_p' is calculated by either using an oedometer test, and using a triaxial pressure test, and using an in-situ test such as SPT, CPT and so on.

Note: If OCR is greater than 1,0 then soil is overconsolidated.
If OCR is equal to 1,0, then soil is normally consolidated.

$$OCR = \frac{\sigma_p'}{\sigma_{vo}'} \begin{cases} > 1 & \text{soil is overconsolidated} \\ = 1 & \text{" " normally consolidated} \end{cases}$$

Determination of σ_p' from an oedometer test;
1) e - logσ' is used.
2) A turning point is selected (The point A)
3) From the point A, a horizontal line (1) is drawn
4) A tangent (2) at the point A is drawn
5) The angle is determined (α)
6) This angle (α) is bisected by a line (3)
7) The straight portion of loading curve is extended (line 4) to meet with line 3
8) The intersection point of line 3 and 4 give us the value of σ_p'



Note! This method was proposed by Casagrande in addition to this method, there are some several methods to calculate the value of σ_p' such as Tavenas's method and Jarlov's method.

Note: OCR (1-2) soil is lightly overconsolidated
OCR (2-5) " " medium "
OCR >5 " " heavily "

The determination of the duration (time) needed for consolidation
The amount of time or duration needed for the completing of consolidation by using a simple equation as shown below.

$$T = \frac{c_v \, t}{H_d^2}$$

c_v: coefficient of consolidation (in m²/year, in cm²/min, in cm²/s)
t: time duration (in year, time)
H_d: drainage path
T: time factor which is dependent on the value of U (degree of consolidation)

for example U = 50 % T = 0,197
U = 90 % T = 0,848

c_v: coefficient of consolidation is calculated from an oedometer test. There are two methods for determining the value of c_v which are → log t method, → $\sqrt{t}$ method

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H_d (m) = drainage length
$H_d = H_0/2$ if double drainage is possible
$H_d = H_0$ " " not possible

$T = \frac{\pi}{4}\left(\frac{U}{100}\right)^2$ $(U \leq 60\%)$

$T = 1.781 - 0.933 \log[100 - U]$ $U > 60\%$

SHEAR STRENGTH

Strength of soils is referred to as the terms "shear strength". It represents the maximal/maximum resistance to any applied load caused by any constraction activity. The equation of shear strength is

- $\tau = c + \sigma \tan\phi$

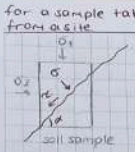
(kPa) τ : strength = shear strength = τ_f [f: failure]
(kPa) c : cohesion [1- if the soil is clay or plastic silt, 2- cohesion can be c', c_d, c_D, c_{cd}, c_{cD}, c_{uu}, c_u, c_{cu}]
(kPa) σ : normal stress [σ normal stress can be σ (total stress) for total stress analysis or short term analysis σ' (effective stress) for effective stress analysis or long term analysis]
(°) ϕ : internal friction angle = shear strength angle = strength angle

ϕ can be
- zero for clay or clay-like soils
- 26°-40° for sand, gravel, sandy gravel, gravelly sand and so on
- 10°-25° for clayey sand, sand silt, sandy clay, silty sand [silt: plastic silt]

Note: If there is no enough time for water to drain out of soil / pore water pressure (u) to be dissipated, and then total stress analysis or short term analysis is carried out. This condition is known as the undrained condition.

yapıray

for a sample taken from a site



soil sample

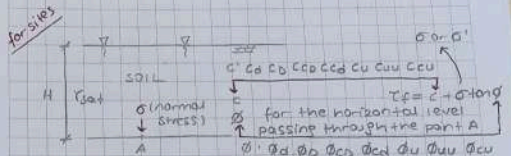
σ_1 : max. principal stress (kPa)
σ_3 : min. " " (kPa)
σ : normal stress acting on the red line (because it is normal to the red line) (kPa) $\left[\frac{\sigma_1+\sigma_3}{2} + \frac{\sigma_1-\sigma_3}{2}\cos 2\alpha\right]$
τ : shear stress acting on the red line $\frac{1}{2}[\sigma_1-\sigma_3]\sin 2\alpha$ (this stress will try to cause the soil sample to fail along the red line)
($^\circ$) α : the angle between the red line and horizontal red line : the slip/failure line for this soil sample

(τ_f, ϕ) (in the laboratory) or $\tau = \frac{1}{2}(\sigma_1 - \sigma_3) \times \sin 2\alpha$

c : c', c_d, c_D — ϕ', ϕ_d, ϕ_D
c_{cd}, c_{cD} — ϕ_{cD}, ϕ_{cd}
c_u, c_{uu}, c_{cu} — ϕ_u, ϕ_{uu}, ϕ_{cu}

$\tau_f = c + \sigma \tan\phi$

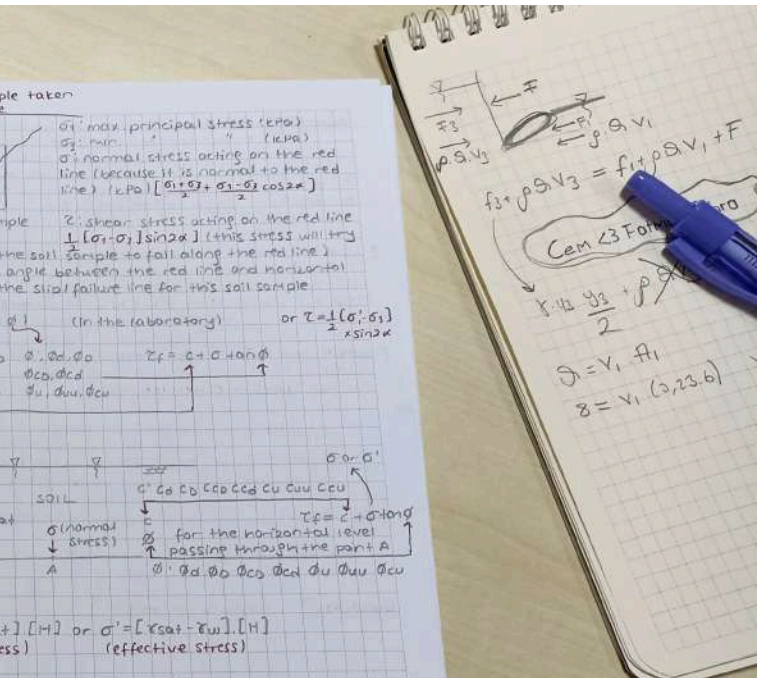
for sites



$\sigma = [\gamma_{sat}] \cdot [H]$ or $\sigma' = [\gamma_{sat} - \gamma_w] \cdot [H]$
(total stress) (effective stress)

Note: If there is enough time for water to drain out of soil pwp to be dissipated, and then effective stress analysis or long term analysis is performed. This condition is known as the drained condition.

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H_d (m) = drainage length
$H_d = H$ / $\frac{H}{2}$ (if double drainage is possible / not possible)

$T = \frac{\pi}{4}\left(\frac{U}{100}\right)^2$ ($U \leq 60\%$)

$T = 1.781 - 0.933 \log(100 - U)$ ($U > 60\%$)

SHEAR STRENGTH

Strength of soils is referred to as the term, "shear strength". It represents the maximal/maximum resistance to any applied load caused by any construction activity. The equation of shear strength is

- $\tau = c + \sigma \tan\phi$

(a)/(a) τ: strength = shear strength = τ_f [f: failure]

(b)/(b) c: cohesion [1- if the soil is clay or plastic silt, 2- cohesion can be c', c_d, c_D, c_{cd}, c_{cD}, c_{uu}, c_u, c_{cu}]

(c)/(a) σ: normal stress [σ normal stress can be σ (total stress) for total stress analysis or short term analysis σ' (effective stress) for effective stress analysis or long term analysis.]

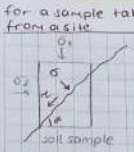
(d)/ϕ: internal friction angle = shear strength angle = strength angle

ϕ can be
- zero for clay or clay-like soils
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Note: If there is not enough time for water to drain out of soil (pore water pressure (u) to be dissipated) and then total stress analysis or short term analysis is carried out. This condition is known as the undrained condition.

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for a sample taken from a site

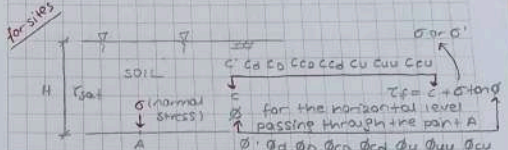


σ_1: max. principal stress (kPa)
σ_3: min. " " " (kPa)
σ: normal stress acting on the red line (because it is normal to the red line) (kPa) $\left[\frac{\sigma_1+\sigma_3}{2} + \frac{\sigma_1-\sigma_3}{2}\cos 2\alpha\right]$
τ: shear stress acting on the red line $\frac{1}{2}[\sigma_1-\sigma_3]\sin 2\alpha$ (this stress will try to cause the soil sample to fall along the red line)
(α): α: the angle between the red line and horizontal red line: the slip/failure line for this soil sample

τ_f, ϕ (in the laboratory) or $\tau = \frac{1}{2}(\sigma_1 - \sigma_3) \times \sin 2\alpha$

c: c_1, c_d, c_D — ϕ: ϕ_d, ϕ_D — $\tau_f = c + \sigma \tan\phi$
c_{cd}, c_{cD} — ϕ_{cD}, ϕ_{cd}
c_u, c_{uu}, c_{cu} — $\phi_u, \phi_{uu}, \phi_{cu}$

for sites



$\sigma = [\gamma_{sat}][H]$ or $\sigma' = [\gamma_{sat} - \gamma_w][H]$
(total stress) (effective stress)

Note: If there is enough time for water to drain out of soil pwp to be dissipated, and then effective stress analysis or long term analysis is performed. This condition is known as the drained condition.

yapıray

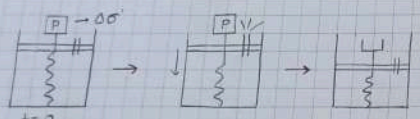
Important! $\phi_u = \phi_{uu} = 0$ (zero) $c' = c_{cd} = c_{cu} = c_{uu} = c_{cd}$ > 0 for normally consolidated clays; > 0 for overconsolidated clays

$\phi = \phi_{cd} = \phi_{cu} = \phi_{uu} = \phi_{cd} > 0$ for all soils

$\sigma_1 = \sigma_1' + u$ $\sigma_2 = \sigma_2' + u$ in other words $\sigma_1' = \sigma_1 - u$ $\sigma_2' = \sigma_2 - u$

! Note 3: For short term or total stress analysis:
→ $\phi_u = 0$ and $c_u > 0$ are used.
For long term or effective stress analysis:
→ $\phi' > 0$ and c' → 1, 2, 3, 4, 5 kPa for normally consolidated
→ > 10 kPa for overconsolidated clays

! Note 4: short term, total stress analysis. undrained condition. long term effective stress analysis are valid for just CLAYS or CLAY-LIKE SOILS.

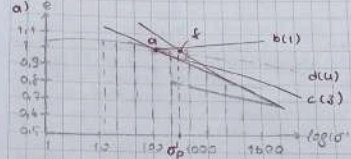


Question ①

Pressure (σ') kN/m²	e (void ratio)
25	1,23
50	1,22
100	0,98
200	0,91
400	0,79
600	0,71
1600	0,62
600	0,635
400	0,655
200	0,67

a) Draw e-log σ' graph and determine the preconsolidation pressure (σ_p' or σ_2').
b) Calculate the compression index and ratio of Cs/Cc
c) Calculate the void ratio at 1200 kPa on the basis of average e-log σ' graph

a)



1) by visual observation establish a point where the radius is minimum
2) draw a horizontal line from point a (ab)
3) draw tangential line the point a
4) draw the line ad and which is bisector of angle abd
5) project the straight portion of the e-log σ' curve back and intersect with the ad line
6) the value of the point f is the preconsolidation pressure.

b) $C_c = \dfrac{e_1 - e_2}{\log\left(\frac{\sigma_2'}{\sigma_1'}\right)}$

	σ' (kPa)	e
1)	200	0,91
2)	400	0,79

$C_c = \dfrac{0,91 - 0,79}{\log\left(\frac{400}{200}\right)} = 0,39$

c) $C_s = \dfrac{e_1 - e_2}{\log\left(\frac{\sigma_2'}{\sigma_1'}\right)}$

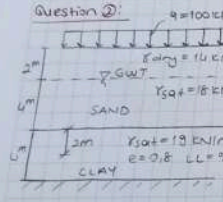
	σ' (kPa)	e
1)	400	0,655
2)	200	0,67

$C_s = 0,049$

yapıray
yapıray

c) $0.39 = \frac{0.8 - e}{\log\left(\frac{1000}{200}\right)} = 0.40$

Question ②:



what is the consolidation settlement of the clay layer if
a) clay is normally consolidated
b) $\sigma'_p = 190$ kPa
c) $\sigma'_p = 190$ kPa
assume that $C_s = \frac{1}{6} C_c$

1) first calculate the effective stress at the middle of the clay layer

$\sigma'_o = 2 \times 14 + 4(18 - 9.81) + 2(18 - 9.81) = 79.14$ kN/m²

2) Find the stress increase in the middle of the clay layer due to the loading. If uniformly load is applied to the very large area $\Delta\sigma' = 100$ kPa
stress increase equal to
stress acting on the soil surface.

$$S_c = \frac{C_c}{1+e_o} \cdot H \cdot \log\left(\frac{\sigma'_o + \Delta\sigma'}{\sigma'_o}\right)$$

$C_c = 0.009\,(LL - 10) = 0.009\,(40 - 10) \rightarrow C_c = 0.27$

$$S_c = \frac{0.27}{1+0.8} \cdot 4 \cdot \log\left(\frac{79.14 + 100}{79.14}\right) = 21.3 \text{ cm}$$

b) $\sigma'_p = 190$ kPa
$\sigma'_o = 79.14 + 100 = 179.14$ kPa

$$S_c = \frac{C_s}{1+e_o} \cdot H \cdot \log\left(\frac{\sigma'_o + \Delta\sigma'}{\sigma'_o}\right) \rightarrow S_c = \frac{0.045}{1+0.8} \cdot 4 \cdot \log\left(\frac{79.14 + 100}{79.14}\right) = 3.6 \text{ cm}$$

$C_s = \frac{1}{6} C_c = \frac{1}{6} \cdot 0.27 = 0.045$

c) $\sigma'_p = 190$ kPa $\quad \sigma'_o = 179.14$ kPa

$$S_c = \frac{C_s}{1+e_o} \cdot H \cdot \log\left(\frac{\sigma'_p}{\sigma'_o}\right) + \frac{C_c}{1+e_o} \cdot H \cdot \log\left(\frac{\sigma'_o + \Delta\sigma'}{\sigma'_o}\right)$$

$$S_c = \frac{0.045}{1+0.8} \cdot 4 \cdot \log\left(\frac{190}{79.14}\right) + \frac{0.27}{1+0.8} \cdot 4 \cdot \log\left(\frac{179.14}{190}\right) = 4.68 \text{ cm}$$

Question ③ According to lab test, the time required for 50 % consolidation of 25mm thick clay is 2min 20sec
a) How long will take the 50 % consolidation of the same clay with 3m thickness, under the same pressure increment?
b) How long will it take for 30 % consolidation in the field?
Note There is a rock layer at the bottom of the clay layer in the field.

a) $U = 50\,\% \longrightarrow T = 0.197$

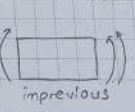
$$C_v = \frac{T \cdot (H_{dr})^2}{t} \qquad 0.025/2$$

$$C_v = \frac{0.197\,(0.0125)^2}{140} = 2.198 \cdot 10^{-7} \text{ m}^2/\text{sec}$$

$$2.198 \cdot 10^{-7} = \frac{0.197\,(3)^2}{t}$$

$t = 8064002 \text{ sec} = 93.33 \text{ day}$

U (%)	T
10	0.008
20	0.031
30	0.0707
40	0.126
50	0.197
60	0.287



yapıray

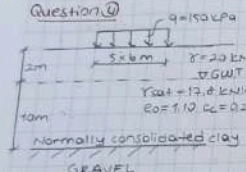
yapıray

b) $U = 30\%$ → $T = 0.0707$

$C_v = \frac{T(Hdr)^2}{t}$ → $2.198 \times 10^{-3} = \frac{0.0707\ (13)^2}{t}$

$t = 33.50$ days

Question ①



Use both → simplified method → assume $\Delta\sigma = 150$ kPa

a) what is the consolidation settlement?

b) The time required for 50% consolidation of a 25 mm thick clay is 12 min. How long will it take for a 10 m thick clay layer in the field to reach 50% conso.?

a) $\sigma'_{vo} = 2 \times 20 + 5(17.8 - 9.81) = 79.95$ kPa

simplified method $\Delta\sigma' = \frac{Q}{(B+z)(L+z)} = \frac{150(5 \times 6)}{(5+7)(6+7)} = 28.846$ kPa

$S_c = \frac{0.25}{1+1.1} \cdot 10 \cdot \log\left(\frac{79.95 + 28.85}{79.95}\right) = 0.16$ m

b) 2nd case $\Delta\sigma' = q = 150$ kPa

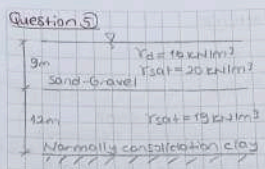
$S_c = \frac{C_c}{1+e_o} \cdot H \cdot \log\left(\frac{\sigma'_o + \Delta\sigma'}{\sigma'_o}\right) = \frac{0.25}{1+1.1} \cdot 10 \cdot \log\left(\frac{79.95+150}{79.95}\right) = 0.54$ m

b) $U = 50\%$ → $T = 0.197$

$C_v = \frac{T(Hdr)^2}{t} = \frac{0.197\ (1.25)^2}{12} = 0.02565$ cm²/min

$0.02565 = \frac{0.197\ (500)^2}{t}$ → $t = 1920077$ min $= 3.66$ years

Question ②



a) what is the consolidation settlement if GWT is lowered 3 m?

b) what is the conso. settlement after 1 year. ($C_v = 2.55 \times 10^{-9}$ m²/sec).

c) $m_v = 0.00018$ m²/kN → coefficient of volumetric compressibility

sr ?

σ'_{vo}	0	25	50	100	200	400	800
e	0.9	0.85	0.87	0.85	0.814	0.775	0.736

a) $\sigma'_{vo} = 3(20 - 9.81) + 6(19 - 9.81) = 144$ Pa 1st

$\sigma'_{vo} = 3 \times 16 + 6.120 - 9.81) + 6.(19 - 9.81) = 156$ kPa 2nd

$\Delta\sigma' = 156 - 144 = 12$ kPa

$C_c = \frac{0.814 - 0.775}{\log\left(\frac{400}{200}\right)} = 0.129$ (?)

$S_c = \frac{C_c}{1+e_o} \cdot H \cdot \log\left(\frac{\sigma'_{vo} + \Delta\sigma'}{\sigma'_{vo}}\right) = \frac{0.129}{1+2.9} \cdot 12 \cdot \log\left(\frac{144+12}{144}\right) = 2.83$ cm

b) $C_v = \frac{T(Hdr)^2}{t}$ → $2.53 \times 10^{-9} = \frac{T.(12)^2}{31536000}$ → $T = 0.593$ → $U = 80\%$

$S = (80\%).(2.83)$

c) $S_c = m_v \cdot H \cdot \Delta\sigma' = 0.00018 \cdot 12 \cdot 12 = 2.736$ cm

yapıray yapıray

SHEAR STRENGTH PARAMETERS

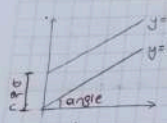
The strength parameters of soils are c and ϕ

SHEAR STRENGTH EQUATION

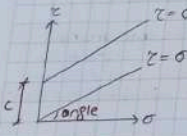
The strength equation of soil is $\tau = c + \sigma \tan\phi$ (for an undrained condition)

This equation can be written by:
- for a drained condition:
  - either $\tau = c + \sigma \tan\phi$ [in terms of total stress]
  - or $\tau = c' + \sigma' \tan\phi'$ [in terms of effective stress]

This equation derived by:
- either $y = ax + b$ (an equation for a straight line)
- or the defination of frictional force which is equal to a frictional coefficient x Force (K x N)



If you construct the same plot using τ and σ instead of y and x



c: cohesion

angle = ϕ



N: the resultant normal force acting on the horizontal AB on which movement will occur

K or Y: frictional coef.

If this blok is thought of as a soil mass and stress are written instead of force

frictional stress = normal stress x frictional coefficient

$\tau = \sigma \times \tan\phi$

N (resultant normal force) becomes σ (normal stress)

K or Y (frictional coefficient) becomes $\tan\phi$ (frictional coefficient for soils)

c and ϕ

The determination of shear strength parameters: The values of cohesion (c) and internal friction angle (ϕ) are obtained from either an in-situ test such as SPT, CPT, In-situ vane, Pressuremeter Test, Plate Loading Test, Dilatometer Test or a laboratory test such as Shear Box Test, Unconfined Compression Test, Triaxial Pressure Test.

The main objective of all the tests is to find c and ϕ

Note 1: Drained Condition: There is enough time for water to drain out of soil / excess pore water pressure to be dissipated within the soil.

Undrained Condition: There is no enough time for " "

Note 2: For sands, the conditions are always drained (earthquake is an expection). For clays, the conditions may be undrained or drained depending on the the rate of loading / the type of engineering activity / the coefficient of k (permeability)

yapıray

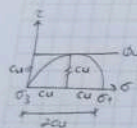
yapıray

Shear Box Test



- for both sands and clays
- no σ_3 no σ_1
- this test performed by drained or undrained
- u can't be measured (pore water pressure)
- $\alpha = 0$ (failure plane is horizontal!)
- duration of this test more or less 5, 10, 20, 30 minutes

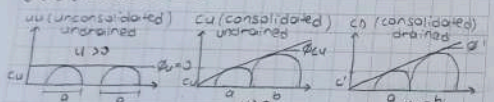
Unconfined Compression Test



- for clays, clay-like soils (plastic silt)
- no σ_3 σ_1 is present
- this test is undrained
- u can't be measured
- $\alpha = 45 + \frac{\phi}{2} = 45°$
- $cu > 0$ $\phi = 0$
- duration of this test more or less 5, 10, 15 minutes

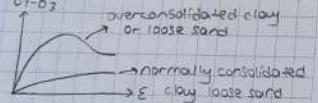
q_u: unconfined compressive strength

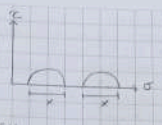
Triaxial Pressure Tests



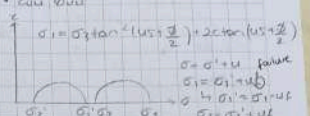
Triaxial Pressure Test (This type of test is carried out in two stages)

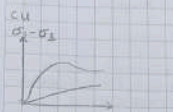
The first stage UU $\sigma_1 - \sigma_3$





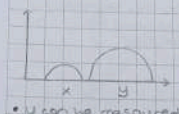
- u can be measured ($u > 0$)
- $\alpha = 45° + \frac{\phi}{2} = 45°$
- c_{uu} ϕ_{uu}



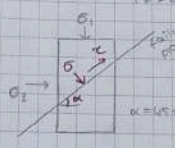


c_{cu} ϕ_{cu} and c' ϕ' are obtained

Note: If the soil is normally consolidated clay
$c_{cu} > 0$, $c' = 0$
If the soil is overconsolidated
$c_{cu} > 0$, $c' > 0$



- u can be measured
- $\alpha = 45 + \frac{\phi}{2}$ ($u > 0$) ($\phi > 0$)



$\alpha = 45 + \frac{\phi}{2}$

σ_1: axial stress pressure (kPa) = max. principal stress
σ_3: minor principal stress (kPa) = confining pressure = cell pressure
$\sigma_1 - \sigma_3$ = deviatoric stress (kPa)

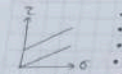
σ and τ are stresses acting on failure plane = failure line / slip plane = slip plane

- Duration of an UU test more or less 5-10-15 minutes $c_u > 0$, $\phi_u = 0$
- " " " " CU " 8, 24, 48, 72 hours depending on the value of k $c_{cu}, \phi_{cu}, c_{co}, \phi_{co}$
- Duration of an CD test 8, 5, 9, 12, 24 months depending on the value of k c_{CD}, ϕ_{CD}

yapıray

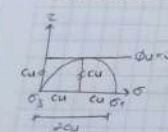
yapıray

Shear Box Test



- for both sands and clays
- no σ_3 no σ_1
- this test performed by drained or undrained
- u can't be measured (pore water pressure)
- $\alpha = 0$ (failure plane is horizontal!)
- duration of this test more or less 5, 10, 20, 30 minutes

Unconfined Compression Test

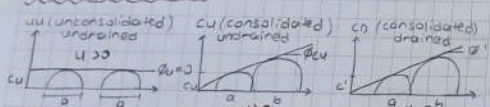


- for clays, clay-like soils (plastic silt)
- no σ_3 σ_1 is present
- this test is undrained
- u can't be measured
- $\alpha = 45 + \frac{\phi}{2} \to 0 = 45°$
- $c_u > 0$, $\phi = 0$
- duration of this test more or less 5, 10, 15 minutes

$c_u = \frac{q_u}{2}$

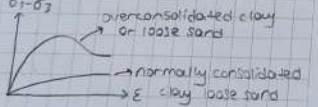
q_u : unconfined compressive strength

Triaxial Pressure Tests



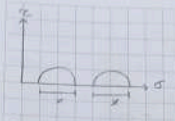
Triaxial Pressure Test (This type of test is carried out in two stages)

The first stage UU $\sigma_3 - \sigma_3$



overconsolidated clay or loose sand

normally consolidated clay, loose sand



- u can be measured ($u > 0$)
- $\alpha = 45 + \frac{\phi}{2}$
- c_{uu} ϕ_{uu}

$\sigma_1 = \sigma_3 \tan^2(45 + \frac{\phi}{2}) + 2c\tan(45 + \frac{\phi}{2})$

$\sigma = \sigma' + u$ failure

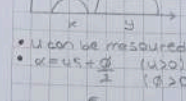
$\sigma_1 = \sigma_1' + u_f$

$\sigma_3 = \sigma_3' + u_f$

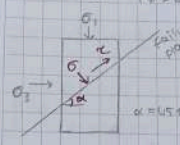
$\hookrightarrow \sigma_3' = \sigma_3 - u_f$

c_{cu} ϕ_{cu} and c' ϕ' are obtained

Note! If the soil is normally consolidated clay
$c_{cu} > 0$, $c' = 0$
If the soil is overconsolidated
$c_{cu} \geq 0$ $c' > 0$



- u can be measured
- $\alpha = 45 + \frac{\phi}{2}$ ($u > 0$) ($\phi > 0$)



σ_1 : axial stress pressure (kPa) = max principal stress

σ_3 : minor principal stress (kPa) = confining pressure = cell pressure

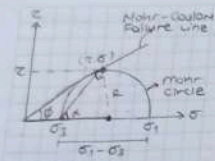
$\sigma_1 - \sigma_3$ = deviatoric stress (kPa)

σ and τ are stresses acting on failure plane = failure line / slip plane = slip lane

* Duration of an UU test more or less 5-10-15 minutes $c_u > 0$, $\phi_u = 0$
* " " " " CU " 8, 24, 48, 72 hours depending on the value of k c_{cu}, ϕ_{cu}, c_{co}, ϕ_{co}
* Duration of an CD test 3, 5, 7, 12, 24 months depending on the value of k c_{co}, ϕ_{co}

yapıray

yapıray

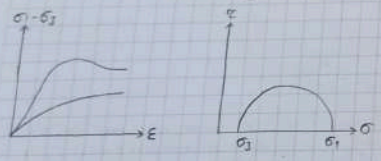


$R = \frac{\sigma_1 - \sigma_3}{2}$

$\sin\phi = \frac{R}{\frac{\sigma_3 + \sigma_1 - \sigma_3}{2}}$

$\sin\phi = \frac{\sigma_1 - \sigma_3/2}{\sigma_1 + \sigma_3/2} = \frac{\sigma_1 - \sigma_3}{\sigma_1 + \sigma_3}$

- The first stage is to apply σ_3. In this stage if the drainage is not allowed, this stage is unconsolidated; is allowed, this stage is consolidated.
- The second stage is to apply σ_1. In this stage if the drainage is not allowed, this stage is undrained allowed, this stage is drained.



- U can be measured ($u = 0$)
- $c' = C_d = C_D = C_{CD}$ are obtained
  $\phi' = \phi_d = \phi = \phi_{CD}$

Note: If the soil is normally consolidated $c' \cong 0$ (generally 0-5 kPa)
If the soil is overconsolidated $c' > 0$ (generally > 10 kPa)

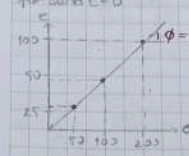
- $\alpha = 45 + \frac{\phi}{2}$.

Question ① For a direct shear test on a dry sand, the following are given; (specimen size: 60 mm × 60mm × 20mm)

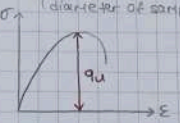
Test No	Normal Force (N)	max. Shear Force (kN)
1	0,48	0,0936
2	0,56	0,1364
3	0,72	0,261

Test No	Normal Stress (kPa)	Shear Stress (kPa)
1	0,48/(0,06)×(0,06) = 50	0,0936/(0,06×0,06) = 26
2	0,361 " " = 100	0,1364 " = 49
3	0,721 " " = 200	0,3611 " = 100

for sand $c = 0$



Question ② An unconfined compression test was conducted on a saturated clay soil. The applied vertical load at failure was Pf = 19 kg, the water content were measured and the height changed were measured 3,85 cm. Calculate the unconfined compressive strength and cohesion of clay soil.
(diameter of sample: 3,95 cm, height of sample: 7,5 cm)



$V_o = A_o H_o = A_f H_f$

$A_f = \frac{A_o H_o}{H_f} \rightarrow \frac{A_o H_o}{H_o - \Delta H} \rightarrow \frac{A_o H_o}{H_o(1 - \frac{\Delta H}{H_f})}$

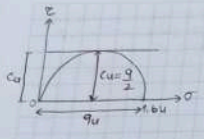
$= \frac{A_o}{1 - \varepsilon_f}$

$\varepsilon_f = \frac{\Delta H}{H_o} = \frac{3,85}{7,5} = 0,05$

$A_f = \frac{A_o}{1 - \varepsilon_f} = \frac{\pi (3,95)^2}{1 - 0,05} = 11,12\ cm^2$

$q_u = \frac{19}{11,12} = 1,64\ kg/cm^2$

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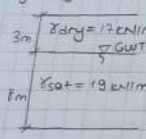


$Cu = \frac{qu}{2} = \frac{1.64}{2} \rightarrow Cu = 0.82\ kp/cm^2$

Question 3) A clay sample was recoverded from the site at 7m depth, Triaxial tests were carried out with these samples. According to the test results,

	(σ_3) Cell Pressure (kPa)	Deviator Pressure (kPa) at failure $(\Delta\sigma_f)$
1. CU	120	197
2. CU	200	329
1. CD	110	131,8
2. CD	200	260

a) Determine the pore water pressures at failure for Cu test
b) Calculate the shear stress of the soil profile at 7m depth by using effective stress parameter



1. CD → $\sigma_3' = 110\ kPa$ $\Delta\sigma_f = 131.8\ kPa$

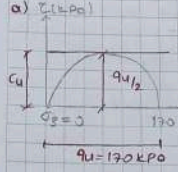
Question 1

Cu: consolidated-drained test
uc: unconfined compression test

	Cell Pressure	Deviator Pressure (kPa)	σ_3' (kPa)
1st CD	150	130	255
2nd CD	200	208	408
3rd UC	—	170	

a) Draw the mohr circle for UC test and calculate the undrained shear strength.
b) Calculate the pore water pressure for CD test
c) Draw the mohr circle for CD test. Determine the drained shear strength parameters.
d) which parameter can be used if the structure is constructed slowly or quickly?

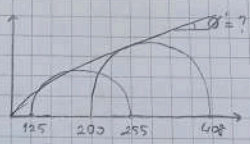
a)



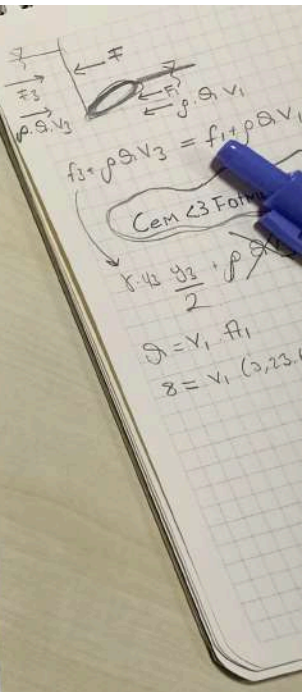
$\tau = c + \sigma \tan\phi$
$Cu = \frac{qu}{2} = \frac{170}{2} = 85\ kPa$

b) 0

c)



$\sigma_3' = \sigma_1' \times \tan^2(45 - \phi'/2)$
$125 = 255 \times \tan^2(45 - \phi'/2)$
$200 = 408 \times \tan^2(45 + \phi'/2)$
$\phi' = 20°$



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Question 2: The equation of the effective stress failure envelope for normally consolidated clayey soil is $\tau_f = \sigma' \tan 30$. A drained triaxial test was conducted with the same soil with a confining pressure at 70 kPa. Calculate the deviator stress at failure.

$c' = 0$
$\tau_f = \sigma' \tan 30°$
$\phi' = 30°$
$\sigma_3 = 70$ kPa

CD → effective parameters
→ $u = 0$

$\sigma_1' = \sigma_3 \times \tan^2(45 + \phi'/2) + 2c\sqrt{\ }$
$\sigma_1' = 70 \times \tan^2(45 + 30/2) \rightarrow \sigma_1' = 210$ kPa
$\Delta\sigma = \sigma_1' - \sigma_3' = 210 - 70 = 140$ kPa

Question 3: The CU test was conducted on a saturated - normally consolidated clay. The sample failed when the deviator stress reached 166 kPa. If the cell pressure during the test is 200 kPa and equation for failure envelope is (effective stress) $\tau_f = \sigma' \times \tan 34$

a) Determine the total strength parameter.
b) Determine the pore water pressure at failure.

$c = 0$ $\quad \sigma_3 = 200$ kPa
$\phi = 34$ $\quad \sigma_1 = 166 + 200 = 366$ kPa
$\sigma_1 = \sigma_3 \times \tan^2(45 + \phi/2) \rightarrow 366 = 200 \tan^2(45 + \phi/2)$
$\rightarrow \phi = 17.05°$ for total

$u_f = ? \rightarrow (366 - u_f) = (200 - u_f) \tan^2(45 + 34/2)$
$u_f = 134.58$ kPa

Question 4: The results of CD test for normally consolidated clay are given below:
cell pressure = 112 kPa (σ_3)
deviator stress at failure = 175 kPa ($\Delta\sigma$)

a) Find the ϕ' ?
b) θ_f
c) σ' and τ' at the failure plane
d) Determine the effective normal stress for max. shear stress

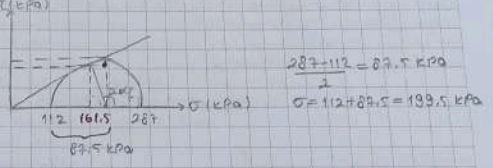
$\sigma_3' = 112$ kPa
$\sigma_1' = \sigma_3' + \Delta\sigma = 112 + 175 = 287$ kN/m²
$\sigma_1' = \sigma_3' \tan^2(45 + \phi/2) + 2c\sqrt{\ }$ → $c = 0$
$287 = 112 \tan^2(45 + \phi/2) \rightarrow \phi' = 26°$

θ_f = angle that the failure plane makes with horizontal plane
$\theta_f = 45 + \phi/2 = 45 + 26/2 = 58°$

$\sigma' = \frac{\sigma_1' + \sigma_3'}{2} + \frac{\sigma_1' - \sigma_3'}{2} \cos 2\theta_f = \frac{287 + 112}{2} + \frac{287 - 112}{2} \cos 2.58$

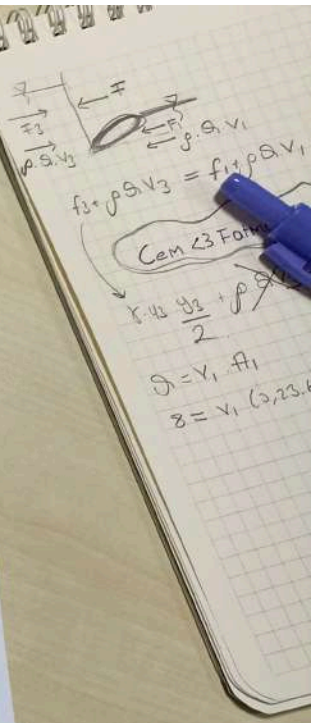
$\sigma' = 161$ kN/m²

$\tau = \frac{\sigma_1' - \sigma_3'}{2} \sin 2\theta_f = \frac{287 - 112}{2} \sin 2.58 \quad \tau = 78.6$ kN/m²



$\frac{287 + 112}{2} = 87.5$ kPa
$\sigma = 112 + 87.5 = 199.5$ kPa

$\sigma' = \frac{287 + 112}{2} + \frac{287 - 112}{2} \cos 90 = 119.50$ kPa

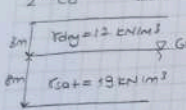


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Question 5:

	Cell Pressure (kPa)	Deviator Pressure (kPa)
1st CU	120	197
2nd CU	200	178
1st CD	110	131,8
2nd CD	200	240



a) Determine the pore water pressure at failure for CU tests.
b) Calculate the shear strength of the soil at 7m depth by using effective stress parameter.

1st CD → $\sigma_3' = 110$ kPa $\sigma_1' = 241,8$ kPa
$\sigma_3' = 200$ kPa $\sigma_1' = 440$ kPa
$\sigma_3' = \sigma_1' \tan^2(45-\phi'/2) - 2c' \tan(45-\phi'/2)$
$-110 = 241,8 \tan^2(45-\phi'/2) - 2c' \tan(45-\phi'/2)$
$200 = 440 \tan^2(45-\phi'/2) - 2c' \tan(45-\phi'/2)$

$\phi_d = 22°$ $c' = 0$

$(120-u_f) = (312-u_f)\tan^2(45-\phi'/2) \rightarrow u_f = 30,7$ kPa
$(200-u_f) = (378-u_f)\tan^2(45-\phi'/2) \rightarrow u_f = 50,5$ kPa

at 7m → $\tau = c' + \sigma' \tan\phi'$
$\sigma' = 3.(17) + 4.(19-9,81) = 87,76$ kPa
$\tau = (87,76)\tan 22 \rightarrow 35,45$ kPa

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COMPACTION

Compaction is a type of soil improvement. Sometimes soil layers on which foundation are built cannot carry loads caused by structures on construction sites. Therefore, the engineering properties of soils need to be improved at these sites. To do so, there are some several methods depending on the type of soil as shown in the table below.

Type of Soil

Methods to be used	(Gravel-sands) Coarse grained soils	(Silts-clay) Fine grained soils
	Compaction Vibrocompaction Vibroflotation Dynamic Compaction Dynamic consolidation	Compaction Pre loading well-points electro-osmosis sand drains
	↓	↓
	the main objective of these methods is to derify the soil by reducing the value of void ratio by vibration. (or is to make the soil into a denser state)	The main objective of these methods is to derify the soil by reducing the value of void ratio by draining water out of the soil.

Important Note: For both types of soil, in addition to the methods stated above,

admixtures
geotextiles
suitable grants for injection

} can be used as different soil improvement techniques

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Compaction is also a kind of compression. But it can't be confused with consolidation which is a kind of compression. These are some significant differences between compaction and consolidation as shown in the table below

Compaction
- the duration is short (15 days, 1 month, 3 month etc.)
- the soil is not saturated ($S \neq 100\%$)
- it occurs under the weight of some vehicles used for completing the process of compression.
  compactors = vehicles : + sheeps foot rollers (for fine grained soil) + vibratory rollers (for coarse grained soils)
- it occurs by sending the air out of the voids of the soil

Consolidation
- the duration is long (5 years, 50 years etc.)
- the soil must be saturated according to the theory of consolidation proposed by Terzaghi
- it occurs under the weight of soil layers and/or structures
- it occurs by sending the water out of the voids of the soil

If the soil layers close to the ground surface do not have the capacity enough to carry superstructure loads, (weak soil), instead of shallow foundation a deep foundation (piled-foundation, pile-foundation) may be preferred (sound strong firm soil). Alternatively, if it is more economical the weak soil can be removed and that area can be filled by using the same material excavated from that site or by using a different material excavated from a different site. The construction of such a fill is actually to carry out the process of compaction. To construct a compacted fill, filling material (soil) are placed in layers (the thickness of each layer is generally between 30 cm - 50 cm) at the site and by using a suitable compactor (vibratory compactors / rollers for coarse-grained soils / sheeps foot compactors / rollers for fine grained soils)

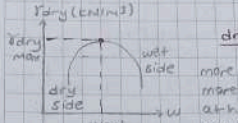
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the fill is constructed. The best soils known in the literature are compacted fills.

If it is more economical, sometimes, instead of constructing a compacted fill by removing the weak soil, that weak soil can be compacted in-place by using a suitable compactor. But in this process, the thickness of soil to be improved is highly limited. (1,5m - 2m at most)

The Factors Affecting Compaction

- water content = moisture content of soil which is obtained from some laboratory tests
- dry unit weight of the soil
  - Standard Proctor Test (25 drops, 2,5 kg, 30,5 cm, 3 layers)
  - modified Proctor Test (25 drops, 4,5 kg, 45 cm, 5 layers)



dry side	wet side
more permeable	less permeable
more compressible at high pressure	more compressible at low pressure
higher strength at wopt	

- Compactor effort = amount of energy = compactor pressure = contact pressure (the load applied from compaction vehicles to the soils)
- number of passes of compaction vehicles (6-10)
- speed of compaction vehicles

Note: wopt: optimum water content at which max dry unit weight is reached.
dry side: the side on which the value of w is less than wopt
wet side: " " " " " " " " more than wopt

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a hole is opened. A soil sample is taken from this hole. And the weight (mass) and the volume of this sample is calculated and γ_{dry} is determined by lab. This γ_{dry} is compared with $\gamma_{dry\,max}$ determined from either standard proctor test or modified " "

by the following equation:

$$RC = \frac{\gamma_{dry\,field}}{\gamma_{dry\,max\,lab}} \times 100$$

(relative compaction)

Important Equations and Concepts For Solving Problems Regarding Compaction

1) $V_T = V_V + V_S \Rightarrow \frac{V_T}{V_S} = \frac{V_V}{V_S} + \frac{V_S}{V_S} \rightarrow V_T = V_S(1+e)$ (e)

$V_{T_{fill}} = V_S(1+e_{fill})$ $\quad V_{T_{exc}} = V_S(1+e_{exc})$

2) $\gamma_{wet} = \frac{\gamma_s}{1+w}$

3) $\gamma_{S_{exc}} \rightarrow \gamma_{S_{fill}} = \gamma_{S_{exc}}$
$W_{S_{exc}} \rightarrow W_{S_{fill}} = W_{S_{exc}}$
$V_{S_{exc}} \rightarrow V_{S_{fill}} = V_{S_{exc}}$

$\gamma_S = \frac{W_S}{V_S}$ $\quad \gamma_S = \frac{M_S}{V_S}$

4) $\gamma_{dry\,fill} \neq \gamma_{dry\,exc}$ $(\gamma_{dry\,fill} > \gamma_{dry\,exc})$

5) $w_{fill} \neq w_{exc}$

6) $V_{fill} \neq V_{exc}$ $(V_{fill} < V_{exc})$

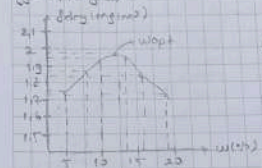
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Question 1

Density No	γ_{wet}	water content (%)	γ_{dry}
1	1,89	5	1,76
2	2,04	8	1,85
3	2,13	10	1,84
4	2,20	12	1,85
5	2,16	14	1,86
6	2,09	19	1,76

For $S(\%) = 80\%$ draw the S curve, determine the optimal water content. Draw the air content curve for $A_v = 10\%$

$\gamma_s = 2,7$ mg/m³



$\gamma_{dry} = \frac{\gamma_{wet}}{1+w}$

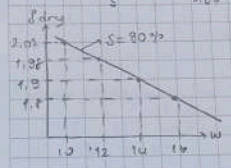
$\gamma_{dry} = \frac{1,87}{1+0,05} = 1,78$

$w_{opt} = 11,8\%$

$\gamma_{dry\,max} = 1,84$ mg/m³

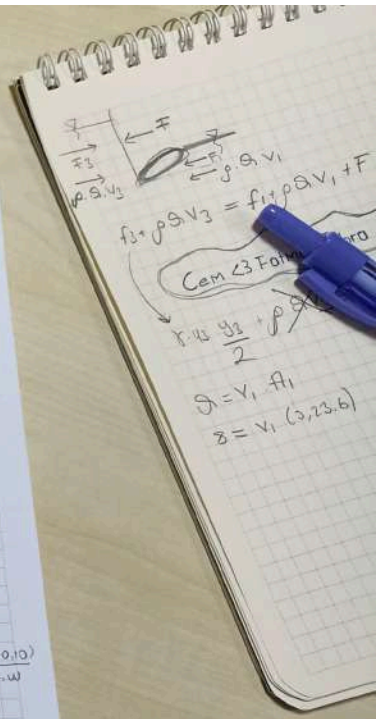
$\gamma_{dry} = \frac{\gamma_s}{1+\frac{\gamma_s \cdot w}{S}} = \frac{2,7}{1+\frac{2,7 \cdot w}{0,80}}$

w 10 %	γ_{dry} 2,02
" 12 %	" 1,88
" 14 %	" 1,9
" 16 %	" 1,8



$\gamma_{dry} = \frac{\gamma_s \cdot (1-A_v)}{1+\gamma_s \cdot w} = \frac{2,7 \cdot (1-0,10)}{1+2,7 \cdot w}$

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Question 2

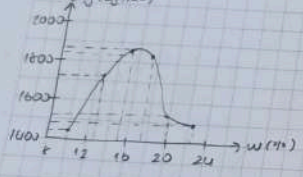
Trial No	mass of moist soil in the mold (kg)	Moisture content w (%)
1	1,42	10
2	1,83	12,5
3	2,02	15
4	1,95	17,5
5	1,73	20
6	1,85	22,5

$\gamma_{dry max}$ = ? w_{opt} = ?

V_{mold} = 943.3 cm^3

$\gamma = \frac{N}{V}$ (kg/m³) $\gamma_{dry} = \frac{\gamma}{1+w}$

1	1558.1	1416.6
2	1940	1724.4
3	2141.4	1862
4	2063.2	1759.3
5	1833.9	1528.2
6	1781.5	1482.4



$\gamma_{dry max}$ = 18.70 kg/m³
w_{opt} = 15 %

For 87 % relative compaction in the field. Field dry density? Range of acceptable moisture content?

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$R(\%) = \frac{\gamma_{dry(field)}}{\gamma_{dry\,max}} \times 100 = \left(\frac{\gamma_{dry\,field}}{18.70}\right) \times 100$

$\gamma_{dry(field)} = (0,87)(18.70) = 16.14$ kg/m³

$w_{ACC.} = 13.5\,\% - 16.5\,\%$

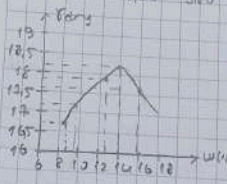
Question 3 An embankment is to be constructed using clayey sand soil. void ratio and specific gravity of soil taken from borrow area are as follows. $e = 0,95$ $G_s = 2,70$ Results of a compaction test conducted on the soil are given below.

a) Plot the compaction curve and determine w_{opt} and $\gamma_{dry max}$

b) Calculate the degree of saturation of a soil compacted with optimum water content.

c) Tests done on the in-situ compacted soil show compaction water content as 12 % and unit weight as 19,6 kN/m³. Calculate the relative compaction (%)

γ_{dry} (kN/m³)	16.62	17.25	17.95	18.10	17.55
w (%)	8.20	9.60	12.50	14.00	16.00



w_{opt} = 14 % $\gamma_{dry max}$ = 18.1 kN/m³

$e = \frac{G_s \cdot \gamma_w}{\gamma_{dry max}} - 1 = \frac{(2,7)(9.81)}{18.1} - 1 = 0,463$

$S \cdot e = w \cdot G_s \rightarrow S = \frac{(0,1)(12,9)}{0,463} = 0,816$

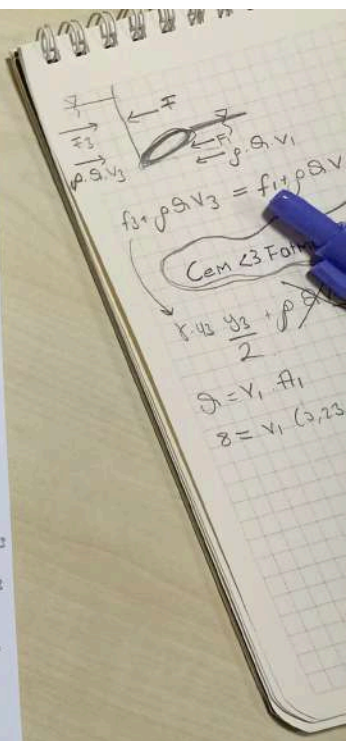
81.6 %

$w = 12\,\%$ $\gamma = 19,6$ kN/m³

$\gamma_{dry} = \frac{\gamma}{1+w} = \frac{19.6}{1+0.12} = 17,5$ kN/m³

$R(\%) = \frac{\gamma_{dry}}{\gamma_{dry\,max}} \times 100 = \frac{17,5}{18,1} \times 100 = 96.7\,\%$

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d) For 30000 m^3 soil taken from borrow area, how much embankment using compacted soil can be made?
$w = 12\,\%$ $\gamma = 19.6\ kN/m^3$

$$\gamma = \frac{G_s \gamma_w (1+w)}{1+e_{emb}} \Rightarrow (1+e) = \frac{G_s \gamma_w (1+w)}{\gamma} \Rightarrow (1+e) = \frac{(2,7)(9,81)(1+0,12)}{19,6} = 1,514$$

$$\frac{V_{TOTAL}}{V_{SOLIDS}} = \frac{1+e_{emb}}{1} \Rightarrow V_T = V_S(1+e_{emb})$$

$$V_S = \frac{V_T}{1+e} = \frac{30000}{1+0,95} = 15385\ m^3$$

$V_{emb} = V_S(1+e_{emb}) \rightarrow V_{emb} = (15385)(1,514) = 23293\ m^3$

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